Multilevel Modeling for Longitudinal Models

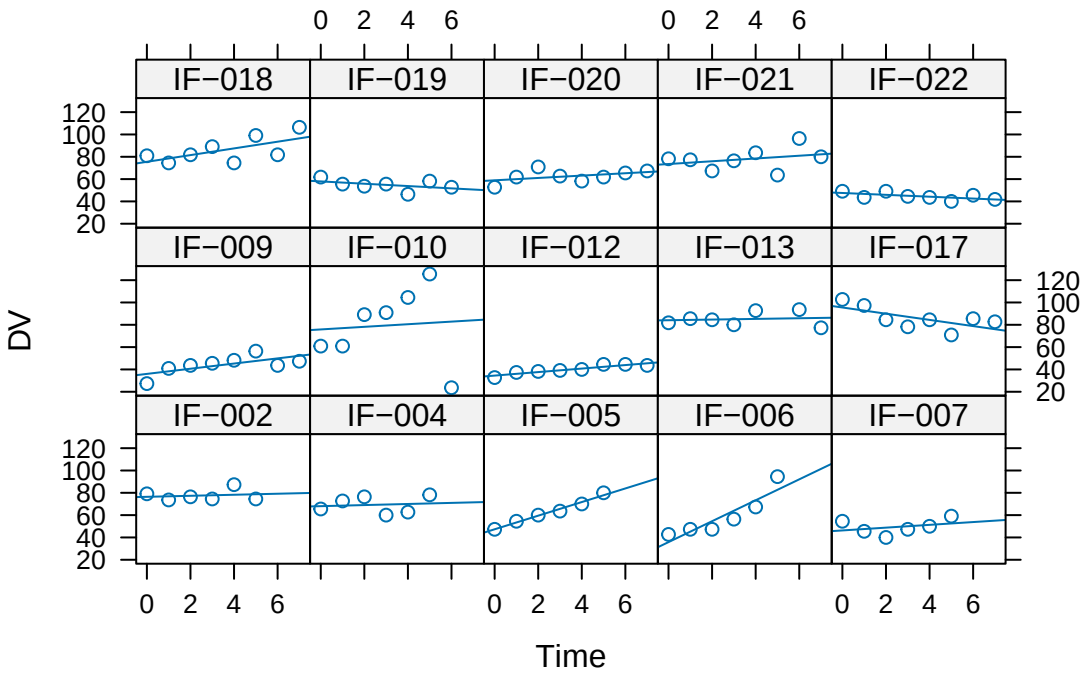
Zhengxiao Wei, Eleah Stringer

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Data description

The longitudinal data are from an experiment to study the effect of intermittent fasting (IF) on the lymphocyte measurements of 15 participants, each of whom was measured before the intervention (one time), during the intervention (either four or six times), and after the intervention (one time).

```
dat <- read.csv("~/Downloads/IF.csv", stringsAsFactors=T)
lattice::xyplot(DV ~ Time | ID, data=dat, type=c("p","r"))
```



The data set (long format) consists of four columns:

- **DV** - [numeric] lymphocyte count ($\times 10^9$/L) as the dependent variable; lower scores correspond to better outcomes;
- **ID** - [factor] participant identifier; 15 levels;
- **Time** - [numeric] time point, with a frequency of every 30 days, starting from Month 0;
- **Trt** - [numeric] treatment identifier; 0 as pretest, 1 as intervention, and 2 as posttest.

The hypothesis is that IF can help patients lower their lymphocyte count.

Linear mixed-effect regression

Using the "nlme" package (v 3.1-162)

```
fit <- nlme::lme(fixed=DV ~ Time + Trt, random=~Time | ID, data=dat)
summary(fit)
```

```
## Linear mixed-effects model fit by REML
##   Data: dat
##        AIC     BIC    logLik
##   873.9152 892.426 -429.9576
##
## Random effects:
##  Formula: ~Time | ID
##  Structure: General positive-definite, Log-Cholesky parametrization
##             StdDev    Corr
## (Intercept) 16.980586 (Intr)
## Time         1.282254 -0.415
## Residual    11.618196
##
## Fixed effects:  DV ~ Time + Trt
##                Value Std.Error DF   t-value p-value
## (Intercept) 62.15588  5.000281 90 12.430477  0.0000
## Time         1.59692  0.921766 90  1.732454  0.0866
## Trt         -1.77026  3.389341 90 -0.522302  0.6027
##  Correlation:
##      (Intr) Time
## Time -0.129
## Trt  -0.267 -0.727
##
## Standardized Within-Group Residuals:
##         Min          Q1         Med          Q3         Max
## -4.92418013 -0.37636877 -0.01469857  0.38091364  3.81650651
##
## Number of Observations: 107
## Number of Groups: 15
```

Using the "lme4" package (v 1.1-33)

```
fit2 <- lme4::lmer(DV ~ Time + Trt + (Time | ID), dat)
summary(fit2) #output omitted
```

In linear mixed-effect regression (LMER), each subject has its intercept and slope (i.e., random intercepts and random slopes). The negative coefficient (-1.77) for the treatment effect indicates that IF may lower the lymphocyte count. However, the p-value of .60 is not statistically significant at an alpha level of .05

```
coef(fit)
```

```
##         (Intercept)       Time       Trt
## IF-002    75.04012 1.29538744 -1.770261
## IF-004    67.08702 1.42333540 -1.770261
## IF-005    58.10572 2.46519600 -1.770261
## IF-006    53.42396 3.06369758 -1.770261
## IF-007    47.77995 1.84648112 -1.770261
## IF-009    39.83458 2.11756666 -1.770261
## IF-010    75.82297 1.42127370 -1.770261
## IF-012    36.76805 1.90434823 -1.770261
## IF-013    81.83512 1.10230413 -1.770261
## IF-017    86.19535 0.07290069 -1.770261
## IF-018    79.42220 1.99564973 -1.770261
## IF-019    53.96410 1.15897586 -1.770261
## IF-020    59.20781 1.54398817 -1.770261
## IF-021    73.68178 1.47717316 -1.770261
## IF-022    44.16939 1.06547352 -1.770261
```